

Effect of Network Depth on Accuracy

Abstract

Neural networks are widely used for image classification and language modelling. In this synthetic study we investigate how network depth affects classification accuracy. Deeper networks often improve accuracy but increase training cost and memory consumption. We evaluate several architectures on a small benchmark dataset and compare their accuracy.

1. Introduction

Neural networks are widely used for image classification and language modelling. In this synthetic study we investigate how network depth affects classification accuracy. Deeper networks often improve accuracy but increase training cost and memory consumption. We evaluate several architectures on a small benchmark dataset and compare their accuracy. Neural networks are widely used for image classification and language modelling. In this synthetic study we investigate how network depth affects classification accuracy. Deeper networks often improve accuracy but increase training cost and memory consumption. We evaluate several architectures on a small benchmark dataset and compare their accuracy.

2. Methodology

Our method trains convolutional networks with different depths using stochastic gradient descent. Each network is trained for fifty epochs and evaluated on a held out validation set. We record accuracy, training time and memory usage for every architecture in the experiment.

3. Results

The deeper network achieved higher accuracy than the shallow network on the validation set. However training time increased substantially as network depth increased. These results suggest that moderate depth gives the best trade off between accuracy and cost.

Model	Depth	Accuracy	Time
Net-A	8	81.2	12
Net-B	16	86.5	25
Net-C	32	88.1	61

Table 1. Accuracy and training time by depth.

4. Discussion

The deeper network achieved higher accuracy than the shallow network on the validation set. However training time increased substantially as network depth increased. These results suggest that moderate depth gives the best trade off between accuracy and cost.

5. Conclusion

We conclude that network depth improves accuracy with diminishing returns. Future work will study regularisation and data augmentation for deep networks. This synthetic document exists only to test the analysis pipeline.